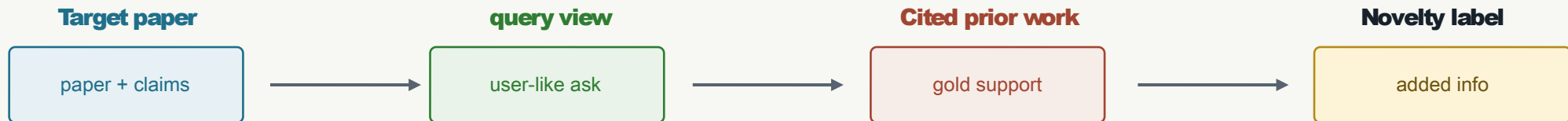


Citation-Graph Benchmark: From Retrieval Labels to Novelty Evidence

Goal: evaluate whether a system can find the prior work that matters for judging what a target paper actually adds.



1. Why citation graph?

Citations are author-supplied evidence of which prior work the paper is positioning against, so they are less circular than using embedding similarity and then asking an LLM to say “good / bad.”

Process it as: target paper -> cited candidate pool -> gold prior support ACUs -> label the target’s added information relative to that support.

Similarity/LLM labels still help rank candidates, but citations anchor the benchmark in real scholarly comparison behavior.

2. How to make queries?

Create several query views from the same target paper: title/abstract, extracted contribution claims, dataset ACUs, and an LLM-rephrased user question.

Use the LLM rephrase only as one query condition, with a prompt such as: “What prior dataset or benchmark should I compare against for this contribution?”

Report whether retrieval is robust across query styles, not just optimized for one phrasing.

3. Non-novel work?

Citation pairs alone will under-sample zero-novelty cases because published papers usually claim some contribution.

Add controls: near-duplicate/repackaging examples, dataset extensions with only format/language splits changed, and hard negatives from highly similar uncited papers.

Calibrate labels into bands: repackaging, incremental, substantial. Then sample deliberately so the benchmark covers the full novelty distribution.

Evaluation framing: retrieve the right prior support first, then score how much of the target’s claims remain unsupported by that prior work.